

Jasreen

Aspiring Software Development Engineer | Machine Learning & Data Analyst

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PROFESSIONAL SUMMARY

ECE-AI undergraduate with hands-on experience in machine learning, deep learning, and software development. Proficient in Python and C++, with strong foundations in Data Structures & Algorithms, OOP, and DBMS. Experienced in building end-to-end ML systems, optimizing models and deploying data-driven applications. Actively seeking SDE / ML Engineer and Data Analyst roles.

WORK EXPERIENCE

Computer Vision & Deep Learning Intern

Jun '25 — Aug '25

Anveshan Foundation, IGDTUW

New Delhi, India

- Engineered and optimized CNN-based deep learning models using PyTorch, TensorFlow, and OpenCV, boosting model inference efficiency by ~20% through architectural refinement and preprocessing improvements.
- Contributed to the implementation of “Multi-Disease Diagnosis from Chest X-Rays”, handling data preprocessing, model training, and evaluation on 112,120 frontal-view X-ray images, from 30,805 patients.
- Improved predictive performance, achieving 89.6% classification accuracy across multiple disease categories through systematic hyperparameter tuning and regularization strategies.

Python ML Intern

Jun '23 — Jul '23

COE-AI, AI Club IGDTUW and Anveshan Foundation

New Delhi, India

- Designed, trained, and validated machine learning models using Scikit-learn, TensorFlow, and Python, realizing up to 15% performance gains via feature engineering and informed model selection.
- Collaborated in peer code reviews and worked with cross-functional teams to convert problem statements into deployable ML solutions.
- Standardized and documented ML pipelines to reduce onboarding time and improve long-term maintainability, and deployed ML models using Streamlit and Flask for interactive inference and evaluation.

PROJECTS AND RESEARCH

Customer Shopping Behavior Analysis [Link](#)

Oct '25 — Dec '25

- Analyzed customer purchasing behavior by cleaning, exploring, and modeling raw consumer data using Python and SQL to uncover customer segments, buying trends, and key business drivers.
- Created an interactive Power BI dashboard and executive-level report that translated analytical insights into actionable strategies for marketing optimization, customer retention, and sales growth.
- Tech Stack:** Python, Pandas, NumPy, PostgreSQL, Power BI

Depression Detection in Code-Mixed Text

Jun '25 — Nov '25

- Implemented an NLP-based depression detection system to identify depressive signals in 10,000+ Hinglish user comments and tweets from YouTube, Twitter, and Reddit, supporting early mental health risk identification.
- Benchmarked ML, DL and transformer-based models, achieving 93.7% accuracy with a CNN-LSTM hybrid through class imbalance handling and robust model evaluation.
- Tech Stack:** Python, TensorFlow, PyTorch, Scikit-learn, Pandas

Movie Recommender System [Link](#)

Jan '25 — Feb '25

- Designed a content-based recommendation engine by preprocessing and vectorizing TMDB metadata (genres, keywords, cast, crew) and computing cosine similarity, enabling accurate and personalized movie suggestions.
- Built an interactive Streamlit web application with TMDB API integration to deliver real-time posters, ratings, and descriptions, showcasing a production-ready system applicable to OTT platforms and personalized content discovery.
- Tech Stack:** Python, Pandas, NumPy, Scikit-learn, Streamlit, TMDB API, Flask

ACADEMIC QUALIFICATIONS

B.Tech in ECE-AI, Indira Gandhi Delhi Technical University for Women (GPA: 7.64/10.00)

Nov '22 — May '26

New Delhi, India

Class XII (CBSE), Banasthali Public School (Percentage: 92.4%)

Apr '21 — Mar '22

New Delhi, India

Class X (CBSE), Banasthali Public School (Percentage: 95.6%)

Apr '19 — Mar '20

New Delhi, India

ACHIEVEMENTS AND CERTIFICATIONS

Computer Vision and Deep Learning Internship | Sep '25

Anveshan Foundation, IGDTUW

Data Analytics Job Simulation - Deloitte Australia | Apr '25

Forage

Python & ML Internship | Jul '23

COE-AI, AI Club IGDTUW and Anveshan Foundation

SKILLS

Programming & Databases: C++, Python, C, R, HTML, CSS, MySQL, PostgreSQL

Frameworks & Libraries: TensorFlow, Pandas, NumPy, Scikit-learn, PyTorch, OpenCV, Keras, Flask

Tools & Platforms: Streamlit, Power BI, MS Excel, Matplotlib, Canva, VS Code, MS PowerPoint

Version Control: Git, GitHub

RELEVANT COURSEWORK

Data Structures & Algorithms, Object Oriented Programming, Computer Networks, DBMS, Artificial Intelligence

PUBLICATIONS

Advanced Dual-Band 27/39 GHz Modified Rectangular Patch Antenna with Inset and Tailored Slots for High-Frequency Ka-Band Applications Feb '25

Published in IEEE Xplore

Designed and simulated an inset-fed patch antenna for high-frequency communication systems.

VOLUNTEERING EXPERIENCE

Education Mentor, Desh Ke Mentor, Delhi Govt.

Jan '23 — Nov '23

Mentored school girls on STEM careers and study strategies.